

$$\text{Min } 3x_1 + 2x_2$$

s.t

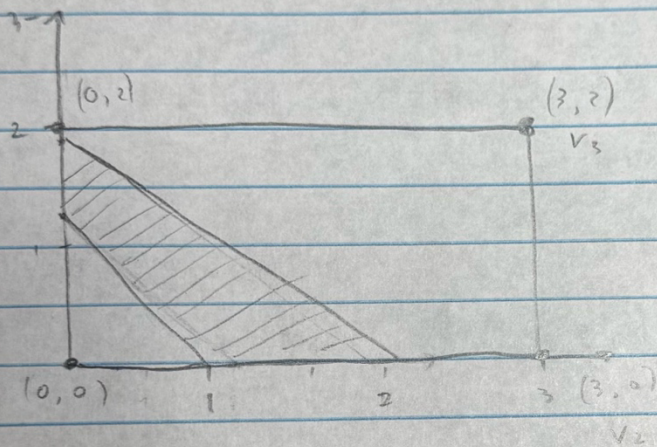
$$5x_1 + 4x_2 \geq 5$$

$$x_1 + x_2 \leq 1.9$$

$$x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



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$$5x_1 + 4x_2 - s_1 + r_1 = 5$$

$$x_1 + x_2 + s_2 = 1.9$$

$$x_1 + s_3 = 3$$

$$x_2 + s_4 = 2$$

$$A_0 = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b_0 = \begin{bmatrix} 5 \\ 1.9 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

RMP

$$\min (3 \ 2) \cdot \sum \lambda_i \cdot v_i$$

s.t

$$\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} \cdot \lambda_i \cdot v_i \geq \begin{bmatrix} 5 \\ 1.9 \end{bmatrix}$$

$$\lambda_i = 1$$

AP

$$\max \left[\pi^T \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

s.t

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Big-M

$$\min M(r_1 + r_2 + r_3)$$

s.t

$$r_1 = s_1 = 5$$

$$r_2 + s_2 = 1.9$$

$$r_3 = 1$$

$$r_1, r_2, r_3 \geq 0$$

$$B = \begin{pmatrix} r_1 & r_2 & r_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad C_B^T (m \ m \ m)$$

$$C_B^T \cdot B^{-1} = (m \ m \ m)$$

$$\text{Max } (m \ m) \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} = (3 \ 2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

s.t.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1, x_2 \geq 0$$

$$\begin{bmatrix} 6m \\ 5m \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow (6m-3)x_1 + (5m-2)x_2$$

Positive in it when

$$x_1 = 3, x_2 = 2 \text{ in MP}$$

$$C_{\lambda_1} = C^T \cdot v_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} (3 \ 2) = 13$$

$$P_{\lambda_1} = \begin{pmatrix} 5 & 4 \\ 1 & 1 \\ \hline & 1 \end{pmatrix} (3 \ 2) = \begin{pmatrix} 23 \\ 5 \\ 1 \end{pmatrix}$$

$$\min \quad M_{r1} + M_{r2} + 13 \lambda_1$$

s.t

$$r_1 + 23\lambda_1 - s_1 = 5$$

$$r_2 + c\lambda_1 + s_2 = 1.9$$

$$r_3 + \lambda_1 = 1$$

$$r_1, r_2, r_3, \lambda_1, s_1, s_2 \geq 0$$

$$\lambda_1 = 1, r_1 = 0, r_2 = 0, r_3 = 0, s_1 = 13, s_2 = 2.9$$

$$B = \begin{pmatrix} \lambda_1 & s_2 & s_1 \\ 23 & -1 & 0 \\ 5 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$C_B^T = (13 \ 0 \ 0)$$

$$C_B^T \cdot B^{-1} = (0 \ 0 \ 13)$$

$$\max \left[(0 \ 0) \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad + \quad 0$$

s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1, x_2 \geq 0$$

$$-3x_1 - 2x_2 + 13 \quad \rightarrow \quad \text{Positive so it enters the MP}$$

$\nearrow x_1 = 0, x_2 = 0$

$$C_{\lambda_2} = C^T \cdot V_2 = 0$$

$$P_{\lambda_2} = \left(\begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\min \quad M(r_1 + r_2 + r_3) + 13\lambda_1 + 0\lambda_2$$

s.t

$$r_1 + 23\lambda_1 + 0\lambda_2 - s_1 = 5$$

$$r_2 + 5\lambda_1 + 0\lambda_2 + s_2 = 19$$

$$r_3 + \lambda_1 + \lambda_2 = 1$$

$$r_1, r_2, r_3 = 0, \quad s_1 = 0, \quad \lambda_1 = 5/23, \quad \lambda_2 = 18/23$$

$$B = \begin{pmatrix} \lambda_1 & \lambda_2 & s_1 \\ 23 & 0 & 0 \\ 5 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$C^T B = (13 \ 0 \ 0)$$

$$C^T B \cdot B^{-1} = (0.56 \ 0 \ 0)$$

$$\max \left[(0.56 \ 0) \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0$$

s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1, x_2 \geq 0$$

$$-0.2x_1 + 0.24x_2 + 0 \rightarrow x_1 = 0, x_2 = 2$$

positive so it enters

MP

$$Ch_2 = C^T \cdot V_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \begin{pmatrix} 0.2 \end{pmatrix} = 4$$

$$P_{22} = \left(\begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right) \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

$$\min \quad M(r_1 + r_2 + r_3) + 13\lambda_1 + 0\lambda_2 + 4\lambda_3$$

s.t

$$r_1 + 23\lambda_1 + 0\lambda_2 + 0\lambda_3 - s_1 = 5$$

$$r_2 + 5\lambda_1 + 0\lambda_2 + 2\lambda_3 + s_2 = 1.9$$

$$r_3 + 1\lambda_1 + 1\lambda_2 + 1\lambda_3 = 1$$

$$B = \begin{matrix} & \lambda_1 & \lambda_2 & \lambda_3 \\ \begin{matrix} 23 & 0 & 0 \\ 5 & 0 & 2 \\ 1 & 1 & 1 \end{matrix} \end{matrix}$$

$$\begin{matrix} 23 & 0 & 0 \\ 5 & 0 & 2 \\ 1 & 1 & 1 \end{matrix}$$

$$\begin{matrix} 5 & 0 & 2 \\ 1 & 1 & 1 \end{matrix}$$

$$\begin{matrix} 1 & 1 & 1 \end{matrix}$$

$$C_B^T = (13 \ 0 \ 0)$$

$$C_B^T \cdot B^{-1} = (-2.3, 13.3, 0)$$

$$\max \left[(-2.3 \ 13.3) \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq 0$$

s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1, x_2 \geq 0$$

$\max = 1.2x_1 + 2.1x_2 \rightarrow$ only positive vertex is already in the MP so this means we've reached the optimal

Phase 2 method

RMP

$$\min 3x_1 + 2x_2$$

s.t

$$\left. \begin{array}{rcl} 5x_1 + 4x_2 - s_1 + r_1 & = & 5 \\ x_1 + x_2 + s_2 & = & 1.9 \\ x_1 + s_3 & = & 3 \\ x_2 + s_4 & = & 2 \end{array} \right\} A_0$$

$$\begin{array}{l} \min C^T \cdot \Sigma \lambda_i v_i \\ \text{s.t} \end{array} \quad \xrightarrow{\text{RMP}} \quad \begin{array}{l} \min r_1 + r_2 + r_3 \\ \text{s.t} \end{array}$$

$$A_0 \Sigma \lambda_i v_i \leq b_0$$

$$\begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} \cdot \lambda_i v_i \leq \begin{bmatrix} 5 \\ 1.9 \end{bmatrix}$$

$$\lambda_i \geq 1$$

AP

$$\begin{array}{l} \max \left[\pi^T \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} - (B^T) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \text{s.t} \end{array}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

1st phase

$$\min r_1 + r_2$$

$$\text{s.t} \quad r_1 - s_1 + s_2 = 5$$

$$r_2 + s_1 = 1.9$$

$$r_2 = 1$$

$$r_1, r_2, s_1, s_2 \geq 0$$

$$B = \begin{matrix} & r_1 & s_2 & r_3 \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

$$C_B^T = (1 \ 0 \ 1)$$

$$C_B^T B^{-1} = (1 \ 0 \ 1)$$

$$\max \left[(1 \ 0) \begin{pmatrix} 5.4 \\ 1.1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0 \rightarrow \max 2x_1 + 2x_2 + 0$$

s.t. → s.t.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$P1_1 = \frac{\begin{pmatrix} 5.4 \\ 1.1 \end{pmatrix} (2 \ 2)}{1} = \begin{pmatrix} 23 \\ 5 \\ 1 \end{pmatrix}$$

$$x_1 = 3 \rightarrow 11 > 0 \rightarrow \text{Enters (rmp!)} \\ x_2 = 2$$

$$\min \quad r_1 + r_2 + r_3 + 0d_1$$

s.t

$$r_1 - s_1 = 0 \quad + 23\lambda_1 = 5$$

$$+ s_2 \quad + 5\lambda_1 = 1.9$$

$$r_2 + +\lambda_1 = 1$$

$$B = \begin{matrix} & \lambda_1 & s_1 & r_2 \\ \begin{pmatrix} 23 & 0 & 0 \\ 5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \end{matrix}$$

$$C_B^T = (0 \ 0 \ 1)$$

$$C_B^T \cdot B^{-1} = [-0.04 \ 0 \ 1]$$

$$\max \left[(-0.04 \ 0 \ 0) \begin{pmatrix} 5.4 \\ 1.1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 1 \rightarrow \max -3.2x_1 + 2.4x_2 + 1$$

$$\text{s.t.} \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\text{s.t.} \quad x_1 \leq 3$$

$$x_2 \leq 2$$

$$P1_2 = \frac{\begin{pmatrix} 5.4 \\ 1.1 \end{pmatrix} (0 \ 0)}{1} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$x_1 = 0 \\ x_2 = 2 \rightarrow 1 > 0 \text{ enters rmp!}$$

now we have a feasible solution without artificial variables composed of $\lambda_1, \lambda_2, s_2$

$$B = \begin{pmatrix} \lambda_1 & s_2 & \lambda_2 \\ 23 & 0 & 0 \\ 5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\lambda_1 = (32) \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 13$$

$$\lambda_2 = (32) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0$$

2nd phase,

RMP

$$\min (32) \cdot \sum \lambda_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda_4 \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

s.t

$$\begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} \cdot \lambda_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda_4 \begin{pmatrix} 3 \\ 0 \end{pmatrix} \leq \begin{pmatrix} 5 \\ 1.4 \end{pmatrix}$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$$

$$\lambda_1, \lambda_2, \lambda_3, \lambda_4 \geq 0$$

AP

$$B = \begin{pmatrix} \lambda_1 & s_2 & \lambda_2 \\ 23 & 0 & 0 \\ 5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$C_B^T = (13 \quad 0 \quad 0)$$

$$C_B^T \cdot B^{-1} = (0.565 \quad 0 \quad 0)$$

max

$$\left[(0.565 \quad 0) \begin{pmatrix} 5 & 4 \\ 1 & 1 \end{pmatrix} - (32) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \max -0.174x_1 + 0.261x_2$$

s.t

→ s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$x_1 \leq 3$$

$$x_2 \leq 2$$

$$x_1 = 0$$

$$x_2 = 2 \rightarrow 0.521 > 0$$

enters Rmp!

$$\lambda_3 = (3 \ 2) \begin{pmatrix} 0 \\ 2 \end{pmatrix} = 4$$

$$P\lambda_3 = \left(\begin{array}{c} (5 \ 4) \ (6 \ 2) \\ \hline 1 \end{array} \right) \geq \begin{array}{c} 0 \\ 2 \\ 1 \end{array}$$

$$B = \begin{pmatrix} \lambda_1 & \lambda_2 & \lambda_3 \\ 2.3 & 0 & 8 \\ 5 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow C_B^T = (1.3 \ 0 \ 8)$$

$$C_B^T B^{-1} = (-2.3 \ 13.3 \ 0)$$

A?

$$\text{Max } \left[(-2.3 \ 13.3) \begin{pmatrix} 5 \ 4 \\ 1 \end{pmatrix} - (3 \ 2) \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + 0$$

s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Max $-1.2x_1 + 2.1x_2 \rightarrow$ Only possible vector to make this positive is already in the basis so this means we have reached the optimal